

Hotel Reservations Cancellation Prediction

Executed project report with saved notebook outputs, tables, and charts

Project Description

Hotel Reservations Cancellation Prediction

Project Summary

This project is a clean, student-friendly implementation of a booking cancellation prediction. The main goal is to predict whether a hotel reservation may be cancelled. The notebook keeps the workflow simple: load the dataset, understand the columns, clean the data, train a baseline model, and check how well the model performs.

No external repository links or profile links are included in this description. Everything needed to understand and run the project is kept inside this folder.

Problem Statement

A raw dataset is useful only when it is converted into a clear decision-making workflow. In this project, the data is processed step by step so that important patterns can be found and a machine learning model can be trained. The expected output is based on noofpreviouscancellations.

Files Included

File	Purpose	Quick Note
Hotel Reservations.csv	Main data source for this folder	2000 sampled rows x 19 columns
Notebook file	Complete Python workflow	Includes loading, cleaning, training, and evaluation
PDF report	Human-readable project summary	Matches the updated project structure

Important Columns

Column	Role
BookingID	Input feature used in the modelling workflow
noofadults	Input feature used in the modelling workflow
noofchildren	Input feature used in the modelling workflow
noofweekendnights	Input feature used in the modelling workflow
noofweeknights	Input feature used in the modelling workflow
typeofmealplan	Input feature used in the modelling workflow
requiredcarparkingspace	Input feature used in the modelling workflow
roomtypereserved	Input feature used in the modelling workflow
leadtime	Input feature used in the modelling workflow
arrivalyear	Input feature used in the modelling workflow
arrivalmonth	Input feature used in the modelling workflow
arrivaldate	Input feature used in the modelling workflow

Workflow Followed

1. Load the data from the local dataset file present in the same project folder.
2. Inspect the structure using shape, column names, missing values, and basic statistics.
3. Clean the dataset by removing empty columns, handling missing values, and keeping only useful features.
4. Prepare the model input by separating features and the target column.
5. Train a baseline model using a simple scikit-learn pipeline with preprocessing.
6. Evaluate the result using practical metrics such as accuracy, classification report, MAE, RMSE, or R2 score depending on the target type.
7. Write observations in a short and direct way so the project can be explained easily.

How to Run

1. Open the notebook in Jupyter Notebook, JupyterLab, Google Colab, or VS Code.
2. Keep the dataset files in the same folder as the notebook.
3. Run the cells from top to bottom.
4. Read the printed metrics and notes at the end before making conclusions.

Final Note

This version keeps the logic understandable and avoids unnecessary complexity. The aim is not to make the notebook look overloaded, but to make it readable, explainable, and useful for a student-level data science portfolio.

Executed Notebook Outputs

The outputs below were generated from the saved notebook after running the code cells in this project folder.

Cell 2 - 2. Load the dataset

```
Dataset shape: (36275, 19)

Booking_ID no_of_adults no_of_children no_of_weekend_nights \
0 INN000001 2 1 0
1 INN000002 3 1 2
2 INN000003 0 0 3
3 INN000004 2 0 1
4 INN000005 1 0 2

no_of_week_nights type_of_meal_plan required_car_parking_space \
0 2 Meal Plan 1 0
1 4 Not Selected 0
2 1 Meal Plan 1 0
3 2 Meal Plan 1 0
4 2 Not Selected 0

room_type_reserved lead_time arrival_year arrival_month arrival_date \
0 Room_Type 1 224 2017 10 2
1 Room_Type 1 4 2018 11 5
2 Room_Type 1 0 2018 2 29
3 Room_Type 1 210 2018 4 19
4 Room_Type 1 47 2018 3 12

market_segment_type repeated_guest no_of_previous_cancellations \
0 Offline 0 1
1 Online 0 1
2 Online 0 0
3 Online 0 0
4 Online 0 0

no_of_previous_bookings_not_canceled avg_price_per_room \
0 1 64.3845
1 1 110.4503
2 1 58.1649
3 1 99.4173
4 1 96.4593

no_of_special_requests booking_status
0 0 Not_Canceled
1 2 Not_Canceled
2 1 Canceled
3 1 Canceled
4 1 Canceled
```

Cell 3 - 3. Understand the data

```
Rows: 36275
Columns: 19

column dtype missing_values \
0 Booking_ID object 0
1 no_of_adults int64 0
2 no_of_children int64 0
3 no_of_weekend_nights int64 0
4 no_of_week_nights int64 0
5 type_of_meal_plan object 0
6 required_car_parking_space int64 0
7 room_type_reserved object 0
8 lead_time int64 0
9 arrival_year int64 0
10 arrival_month int64 0
11 arrival_date int64 0
12 market_segment_type object 0
13 repeated_guest int64 0
14 no_of_previous_cancellations int64 0
15 no_of_previous_bookings_not_canceled int64 0
16 avg_price_per_room float64 0
17 no_of_special_requests int64 0
18 booking_status object 0

unique_values
0 36275
1 6
2 7
3 8
4 19
5 4
6 2
7 7
8 372
9 2
10 14
11 33
12 5
13 2
14 12
15 52
```

```
16 35126
17 7
18 2
```

Cell 4 - 4. Clean the data

Shape after simple cleaning: (36275, 19)

```
Booking_ID no_of_adults no_of_children no_of_weekend_nights \
0 INN00001 2 1 0
1 INN00002 3 1 2
2 INN00003 0 0 3
3 INN00004 2 0 1
4 INN00005 1 0 2

no_of_week_nights type_of_meal_plan required_car_parking_space \
0 2 Meal Plan 1 0
1 4 Not Selected 0
2 1 Meal Plan 1 0
3 2 Meal Plan 1 0
4 2 Not Selected 0

room_type_reserved lead_time arrival_year arrival_month arrival_date \
0 Room_Type 1 224 2017 10 2
1 Room_Type 1 4 2018 11 5
2 Room_Type 1 0 2018 2 29
3 Room_Type 1 210 2018 4 19
4 Room_Type 1 47 2018 3 12

market_segment_type repeated_guest no_of_previous_cancellations \
0 Offline 0 1
1 Online 0 1
2 Online 0 0
3 Online 0 0
4 Online 0 0

no_of_previous_bookings_not_canceled avg_price_per_room \
0 1 64.3845
1 1 110.4503
2 1 58.1649
3 1 99.4173
4 1 96.4593

no_of_special_requests booking_status
0 0 Not_Canceled
1 2 Not_Canceled
2 1 Canceled
3 1 Canceled
4 1 Canceled
```

Cell 5 - 5. Select the target column

Selected target column: no_of_previous_cancellations

```
0 1
1 1
2 0
3 0
4 0
Name: no_of_previous_cancellations, dtype: int64
```

Cell 6 - 6. Quick EDA

Numeric columns: 14
Categorical columns: 5
No missing values found in the previewed columns.

```
count mean std min \
no_of_adults 36275.0 1.840827 0.963137 0.0
no_of_children 36275.0 0.414307 0.598752 0.0
no_of_weekend_nights 36275.0 0.966203 0.994895 0.0
no_of_week_nights 36275.0 2.231261 1.595475 0.0
required_car_parking_space 36275.0 0.030986 0.173281 0.0
lead_time 36275.0 85.240799 85.918748 0.0
arrival_year 36275.0 2017.820427 0.383836 2017.0
arrival_month 36275.0 7.422826 3.169117 0.0
arrival_date 36275.0 15.596361 8.774546 0.0
repeated_guest 36275.0 0.025637 0.158053 0.0
no_of_previous_cancellations 36275.0 0.353329 0.588723 0.0
no_of_previous_bookings_not_canceled 36275.0 0.480110 1.799009 0.0
avg_price_per_room 36275.0 103.439667 35.163452 0.0
no_of_special_requests 36275.0 0.797216 0.913656 0.0

25% 50% 75% \
no_of_adults 1.00000 2.0000 3.0000
no_of_children 0.00000 0.0000 1.0000
no_of_weekend_nights 0.00000 1.0000 2.0000
no_of_week_nights 1.00000 2.0000 3.0000
required_car_parking_space 0.00000 0.0000 0.0000
lead_time 17.00000 57.0000 126.0000
arrival_year 2018.00000 2018.0000 2018.0000
arrival_month 5.00000 8.0000 10.0000
arrival_date 8.00000 16.0000 23.0000
repeated_guest 0.00000 0.0000 0.0000
no_of_previous_cancellations 0.00000 0.0000 1.0000
```

```
no_of_previous_bookings_not_canceled 0.00000 0.0000 1.0000
avg_price_per_room 80.49795 99.1234 121.1754
no_of_special_requests 0.00000 1.0000 1.0000
```

```
max
no_of_adults 5.0000
no_of_children 9.0000
no_of_weekend_nights 7.0000
no_of_week_nights 18.0000
required_car_parking_space 1.0000
lead_time 444.0000
arrival_year 2018.0000
arrival_month 13.0000
arrival_date 32.0000
repeated_guest 1.0000
no_of_previous_cancellations 14.0000
no_of_previous_bookings_not_canceled 58.0000
avg_price_per_room 528.9805
no_of_special_requests 6.0000
```

Cell 7 - 7. Prepare features and target

```
Dropped ID-like columns: ['Booking_ID', 'avg_price_per_room']
Using 1200 sampled rows for model training speed.
Target had rare numeric classes, so it is handled as regression.
Problem type: regression
Feature shape: (1200, 16)
Target unique values: 5
```

Cell 8 - 8. Train a baseline model

Training completed.

Cell 9 - 9. Evaluate the model

```
MAE : 0.4255
RMSE: 0.5685
R2 : -0.1702
```

